Ali Arbab

Class XII student who builds and runs production software. Founder and developer of StockSaathi, a free stock market simulator for students in India that placed 2nd nationally at the Masters' Union AI Buildathon and has grown by word of mouth — plus SpendInCheck, BolHisaab, MagLock Protocol, Sovereign Alpha and LameCRAFT.

CONTACT
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EDUCATION

Class XII (CBSE) — final year

→ Class XII (CBSE): Physics · Chemistry · Mathematics · Computer Science

→ Self-directed: behavioural finance, Postgres security (RLS and grants), embedded power design, evaluating LLMs from production logs

PROJECTS

StockSaathi — A free stock market simulator that teaches students in India how markets — and their own reactions to them — actually work

LIVE IN PRODUCTION · 2026

Vanilla ES modules, no build step · Python on Vercel (one ASGI entrypoint) ·
Supabase Postgres + RLS + RPCs · LLMs on Vertex AI, with fallbacks ·
GitHub Actions data + order jobs · Yahoo + Tickertape + AMFI feeds · Service Worker PWA ·
Kotlin + Jetpack Compose (Android port)

Live: stocksaathi.co.in · Source: github.com/thealiarbab/StockSaathi

SpendInCheck — A personal finance tracker that tells you whether you're on budget, not just what you spent

LIVE — OPEN DEMO · 2026

React 19 + TypeScript + Vite · TanStack Query + React Router · Flask JSON API ·
PostgreSQL (Supabase), raw SQL · psycopg2 connection pool · pytest against a real database ·
StockSaathi public API for prices

Live: spendincheck.com · Source: github.com/thealiarbab/SpendInCheck

BolHisaab — A voice-first Hindi and Hinglish khaata for the 63M shopkeepers still running paper books

REDESIGN + KOTLIN REWRITE · 2026

Next.js 16 + React 19 · TypeScript strict · Tailwind CSS v4 · Supabase + anonymous auth + RLS ·
Llama 3.1 8B (Groq) primary · Llama 3.3 70B fallback · Sarvam Saarika v2 STT ·
Sarvam Bulbul v2 TTS · Whisper-large-v3-turbo fallback · Kotlin (rewrite in progress)

Source private until the rewrite ships

MagLock Protocol — A two-door ESP32 + Flutter smart lock that runs on the home network, with remote access only through a private tunnel

BEING REBUILT FROM SCRATCH · 2026

ESP32 (Arduino, ArduinoJson) · ESP32-CAM (hardware JPEG, MJPEG) · Flutter (Dart 3, Material 3) · Provider state · Cloudflare Tunnel (remote access) · speech_to_text + flutter_tts · Grok-3 (Maggy's reasoning) · Inno Setup Windows installer

Source publishes with the rebuild

Sovereign Alpha — A deterministic, fully local simulation of 2015–2025 markets for testing whether an LLM reading history can find real alpha

RESEARCH — BUILT AHEAD OF HARDWARE · 2026

Python 3.12 + uv · Polars (as_of_join only) · Pydantic + Pandera contracts · Hypothesis property tests · ZeroMQ + MessagePack bridge · Unreal Engine 5 (twin, planned) · DeepSeek-R1 32B (planned)

Source: github.com/thealiarbab/sovereign-alpha

LameCRAFT — A self-hosted home server, the control panel that runs it, and the design system everything else borrows

LIVE — SELF-HOSTED · 2026

FastAPI + uvicorn · Cloudflare Tunnel · Python control panel (psutil) · PyInstaller Windows build · Command Nexus design system (JSON spec) · Standalone HTML + KaTeX pages

Live: lamecraft.org · Source not public

ACTIVITIES & LEADERSHIP

Builder — SpendInCheck 2026-09-PRESENT

Turned my Class XII Computer Science practical into a live personal-finance product: a React client over a Flask and Postgres API, with holdings priced live from StockSaathi.

Founder + sole developer — StockSaathi 2026-04-PRESENT

Built and run a free stock market simulator for students in India: real NSE and BSE prices, nine behavioural-bias detectors and an AI coach that never gives tips. Grown by word of mouth.

Founder + sole developer — BolHisaab 2026-04-PRESENT

Voice-first Hindi and Hinglish ledger for shopkeepers. The web prototype is complete; the app is now being redesigned and rewritten natively in Kotlin.

Builder — MagLock Protocol smart lock 2026-04-PRESENT

Two-door ESP32 smart lock with an ESP32-CAM and a Flutter app, reachable from outside only through a tunnel I control. Being rebuilt from the ground up.

Independent research — Sovereign Alpha 2026-04-PRESENT

Designing and building a fully local, reproducible market simulation to test whether a language model reading history can find real, friction-adjusted alpha.

Builder — national AI buildathon team 2026-04-2026-04

Built StockSaathi end to end during a 36-hour national buildathon while a teammate pitched it to the panel. It placed second nationally.

Field research — local shopkeepers 2026-04-PRESENT

Interviewed shopkeepers about how they actually keep their books before designing BolHisaab — credit, cash, customers named in Hindi, and a paper notebook.

Self-hosting — LameCRAFT home server 2026-02-PRESENT

Run my own server for my sites and tools behind a Cloudflare tunnel, with a Python control panel and the Command Nexus design system my other projects borrow.

Growth — StockSaathi brand ambassador 2026

Devansh was already shouting StockSaathi's name everywhere, unasked, at the risk of his discipline-coordinator badge, so I made it official. Every sign-up since has come by word of mouth.

Robotics — bionic hand exoskeleton 2025-12-PRESENT

Built a biomimetic bionic hand and hand-exoskeleton prototype as part of a robotics group project; the hand was shown at a science exhibition.

Peer programming mentor 2024-PRESENT

Teaching a younger neighbour, now in Class X: programming, hosting servers, how packets and networks work. After a two-day rescue of his locked-out PC, he conceded I'm a "PC goat".

AWARDS & RECOGNITION

2nd place nationally — AI Buildathon MASTERS' UNION · 2026

The brief was set on the day, so StockSaathi was built from scratch in 36 hours; a teammate pitched it to the panel.

Inter-House Math Quiz — 1st place INTER-HOUSE COMPETITION · 2025

Three tie-breakers deep, I had made my peace with third place — then my house was announced first. December 2025.

SKILLS

LANGUAGES

TypeScript · JavaScript · Python · Kotlin · SQL + PL/pgSQL · Dart · C++ (Arduino) · HTML/CSS

FRAMEWORKS & RUNTIMES

Next.js 15 / 16 · React 19 · Tailwind v4 · Jetpack Compose · Flutter · Flask · FastAPI · Vite

DATA & BACKEND

PostgreSQL (Supabase) · Row-level security + grants · SECURITY DEFINER RPCs · Vercel serverless + Edge · Scheduled jobs (GitHub Actions)

AI

LLMs on Vertex AI · Groq (Llama 3.x, Whisper) · Sarvam speech (STT + TTS) · Tool-calling agents · Evaluating from production logs

HARDWARE

ESP32 firmware · ESP32-CAM MJPEG streaming · Fail-secure relay control · Power budgeting (coin cell + LiFePO4) · LoRa mesh (design)

INFRA & TOOLING

Vercel · Cloudflare Tunnel · GitHub Actions CI · Vitest + pytest · Sentry · Husky + lint-staged